UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 005 · Homogeneous functions and Euler's theorem · compiled 14 October 2026

Contents

1. Short notes
2. Keyword index
3. Rapid recall
4. Last 10 minutes

1 Short notes

1 · Homogeneous functions

$u = f(x, y)$ is homogeneous of degree n if $u = x^n g(y/x)$ or $y^n g(x/y)$; in three variables $x^n g(y/x, z/x)$. Equivalently $u(tx, ty) = t^n u$. A polynomial is homogeneous if all its terms have the same degree. The degree may be positive, negative, fractional or zero, e.g. $(\sqrt{x} + \sqrt{y})/(x + y)$ has degree $-\frac{1}{2}$.

2 · Euler's theorem

Euler's theorem: if $u = f(x, y)$ is homogeneous of degree n then $xu_x + yu_y = nu$; for $f(x, y, z)$, $xu_x + yu_y + zu_z = nu$. So only homogeneity and the degree are needed, not the derivatives. Proof: differentiate $u(tx, ty) = t^n u$ with respect to t and put $t = 1$.

3 · Second and higher order: the corollary

Corollary: $x^2 u_{xx} + 2xyu_{xy} + y^2 u_{yy} = n(n-1)u$; adding the first-order result gives $n^2 u$. For a sum of homogeneous parts of different degrees apply this to each part, e.g. degrees 3 and 4 give $9u_1 + 16u_2$. Third order: $n(n-1)(n-2)u$.

4 · Extension: when f(u) is homogeneous

If $f(u)$ (not u) is homogeneous of degree n: $xu_x + yu_y = g(u) = nf(u)/f'(u)$ and $x^2 u_{xx} + 2xyu_{xy} + y^2 u_{yy} = g(u)[g'(u) - 1]$. For $u = \tan^{-1}\frac{x^4+y^4}{xy}$: $f = \tan$, $n = 2$, $g = \sin 2u$, second order $\sin 2u(1 - 4\sin^2 u)$.

5 · Applications of homogeneous functions

Constants are homogeneous of degree 0 and linear functions of degree 1. Euler's theorem gives the scaling of a function; it is used in economics (production functions, marginal productivity, elasticity) and in thermodynamics (extensive variables degree 1, intensive degree 0). A function of ratios has degree 0, so its Euler sum is 0.

2 Keyword index

20 terms; the number in brackets is the topic.

Arbitrary function (1). The g in $x^n g(y/x)$: any function of the ratios.

Constant function (5). Homogeneous of degree 0.

Corollary (Euler) (3). $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$.

Degree (1). The n in $x^n g(y/x)$.

Degree 0 (5). Depends on ratios only; Euler sum 0 (not necessarily constant).

Economics (5). Production functions, marginal productivity, elasticity of demand.

Euler's theorem (2). $xu_x + yu_y = nu$ for homogeneous u of degree n.

Extension of Euler's theorem (4). $xu_x + yu_y = nf(u)/f'(u)$ when $f(u)$ is homogeneous.

Extensive variable (5). Scales with the system: degree 1 (mass, volume).

Homogeneous differential equation (1). Solved with $y = vx$; uses the same idea.

Homogeneous function (1). $u(tx, ty) = t^n u$.

Homogeneous polynomial (1). All terms of the same total degree.

Intensive variable (5). Unchanged by scaling: degree 0 (pressure, temperature).

Linear function (5). Homogeneous of degree 1 (no constant term).

Mixed partial (3). $u_{xy} = u_{yx}$, giving the $2xy$ term.

n(n−1)u (3). Right side of the second-order corollary.

Ratio (1). y/x, z/x: the variables of g.

Scaling (1). Multiplying all variables by t; multiplies u by t^n.

Sum of homogeneous parts (3). Apply Euler to each part with its own degree.

Third-order Euler (3). $\sum$ with coefficients 1, 3, 3, 1 $= n(n-1)(n-2)u$.

3 Rapid recall

1	Define a homogeneous function of degree n.	$u = x^n g(y/x)$ or $y^n g(x/y)$.
2	Scaling form of the definition?	$u(tx, ty) = t^n u(x, y)$.
3	Degree of $3x + 4y$?	1.
4	Degree of $x^3 + y^3 + z^3 + 3xyz$?	3.
5	Degree of $(\sqrt{x} + \sqrt{y})/(x + y)$?	$-\frac{1}{2}$.
6	Degree of a constant?	0.
7	Is every degree-0 function constant?	No: y/x has degree 0.
8	Euler's theorem (two variables)?	$xu_x + yu_y = nu$.
9	Euler's theorem (three variables)?	$xu_x + yu_y + zu_z = nu$.
10	Second-order corollary?	$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$.
11	First plus second order?	n^2u.
12	Third order right side?	$n(n-1)(n-2)u$.
13	$u = x^3y^2 \sin^{-1}(y/x)$: degree, Euler sums?	5; $5u$ and $20u$.
14	Extension: $xu_x + yu_y$ when $f(u)$ is homogeneous?	$g(u) = nf(u)/f'(u)$.
15	Extension, second order?	$g(u)[g'(u) - 1]$.
16	$u = \tan^{-1}\frac{x^4+y^4}{xy}$: $xu_x + yu_y$?	$\sin 2u$.
17	Same u: second order?	$\sin 2u(1 - 4\sin^2 u)$.
18	$x^3 \sin^{-1}\frac{y}{x} + y^{-3}\tan^{-1}\frac{x}{y}$: first + second order?	$9u$.
19	$x^3 \sin^{-1}\frac{y}{x} + x^4\tan^{-1}\frac{y}{x}$ at $(1, 1)$?	$17\pi/2$.
20	Extensive / intensive degrees?	1 / 0.

Mnemonic (study aid). *"Order k brings the falling product."* First order n, second $n(n-1)$, third $n(n-1)(n-2)$; the coefficients follow the binomial row 1, 2, 1 or 1, 3, 3, 1.

4 Last 10 minutes before the exam

Top facts

1. Homogeneous of degree n: $u = x^n g(y/x)$, equivalently $u(tx, ty) = t^n u$.
2. Degrees add for products and subtract for quotients; any degree is allowed.
3. Euler: $xu_x + yu_y = nu$ (three variables: add zu_z).
4. Corollary: $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$; with the first order added, n^2u.
5. Sums of parts with different degrees: apply each degree to its own part.
6. Extension: if $f(u)$ has degree n, $xu_x + yu_y = g = nf/f'$ and the second order is $g(g' - 1)$.
7. $\tan^{-1}(\cdot)$ gives $g = \frac{n}{2}\sin 2u$; $\sin^{-1}(\cdot)$ gives $g = n\tan u$.
8. Degree 0 means a function of ratios only, so Euler's sum is 0; it does not mean constant.
9. ESA favourites: $5u$ & $20u$; $\sin 2u$ & $\sin 2u(1 - 4\sin^2 u)$; $9u$; $17\pi/2$.
10. Applications: scaling, economics, thermodynamics (extensive degree 1, intensive degree 0).

Top formulas

E1.1 · Homogeneous of degree n

ESA ★★

$$u = x^n g\left(\frac{y}{x}\right) = y^n h\left(\frac{x}{y}\right) \quad \Longleftrightarrow \quad u(tx, ty) = t^n\, u(x, y)$$

GIVES	The definition and the equivalent scaling test.
APPLIES	Two variables; with three, use $x^n g(y/x, z/x)$.
NOTE	The scaling test is added for clarity; the slides use the $x^n g(y/x)$ form.

S005·3 · S005·4 · S005·5

E2.1 · Euler's theorem, two variables

ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u$$

GIVES	The scaled sum of first partials.
APPLIES	u homogeneous of degree n.
NOTE	Compact form: $xu_x + yu_y = nu$.

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E3.1 · Corollary: second order

ESA ★★★

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)\,u$$

GIVES The scaled sum of second partials.

APPLIES u homogeneous of degree n with continuous second partials.

NOTE Coefficients $1, 2, 1$ as in $(x+y)^2$.

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E3.3 · First plus second order

ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} + x\,u_x + y\,u_y = n(n-1)u + nu = n^2u$$

GIVES The combination asked in ESA Jan 2024 sets 3 and 4.

APPLIES u homogeneous of degree n; for $u = u_1 + u_2$ apply to each part.

NOTE Degrees 3 and -3 both give 9: so $9u$ for $x^3\sin^{-1}\frac{y}{x} + y^{-3}\tan^{-1}\frac{x}{y}$.

ADDED FOR CLARITY ESA Jan-24 s3 1c · ESA Jan-24 s4 1d · CW U1 P23

E4.1 · Extension, first order

ESA ★★★

$$f(u)\text{ homogeneous of degree }n \implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)} = g(u)$$

GIVES Euler's combination for a function that is not itself homogeneous.

APPLIES Some $f(u)$ is homogeneous (e.g. $\tan u$, $\sin u$, e^u).

NOTE $\tan u$: $g = n\tan u\cos^2 u = \frac{n}{2}\sin 2u$.

S005·7

E4.2 · Extension, second order

ESA ★★★

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)\left[g'(u) - 1\right]$$

GIVES The second-order combination for the extension.

APPLIES $g(u) = n\,f(u)/f'(u)$ from E4.1.

NOTE Differentiate g with respect to u.

S005·7